

PSA

◦ Assignments:

□ Matlab 5 due on Sunday, July 3.

(Run each section separately)

□ Homework 5 due on Tuesday, July 5

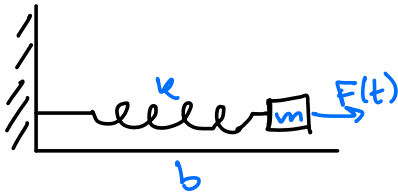
Choose 5 out of 6.

Lecture 11 4.4 Forced Harmonic Oscillator (continued)

5.1. Equilibrium Point Analysis

5.2 Qualitative Analysis

Recap.



$$m\ddot{x} + b\dot{x} + kx = F(t)$$

Standard form:

$$\ddot{x} + p\dot{x} + qx = f(t)$$

extended linearity principle:

$$y(t) = y_h(t) + y_p(t)$$

$$y_h'' + py_h' + qy_h = 0$$

$$y_p'' + p \cdot y_p' + qy_p = f(t)$$

types of forcing: e^{at} , $t^3 + t + 1$, $\cos(\omega t)$, $\sin(\omega t)$

• A few phase portraits for $y'' + 2y' + 10y = 4\cos(2t)$.

Undamped case: $p=0$

$$y'' + \omega_0^2 y = \cos(\omega t)$$

- beating ω close to ω_0
- resonance $\omega = \omega_0$

Ex Determine the frequency of the beats, the rapid oscillations and sketch a solution.

$$\lambda^2 + 4 = 0$$

$$\lambda = \pm 2i$$

$$y'' + 4y = \cos\left(\frac{9}{4}t\right)$$

Step 1: $y_h(t) = c_1 \cos(2t) + c_2 \sin(2t)$

Step 2: $y_p(t) = \frac{-16}{17} \cos\left(\frac{9}{4}t\right)$

a) $\frac{-16}{17} \cos\left(\frac{9}{4}t\right)$ b) $\frac{16}{17} \cos\left(\frac{9}{4}t\right)$ poll 1.

c) $\frac{-16}{17} \sin\left(\frac{9}{4}t\right)$ d) $\frac{16}{17} \sin\left(\frac{9}{4}t\right)$

$$y_c = a e^{i\frac{9}{4}t}, y_p = A \cos\left(\frac{9}{4}t\right) + B \sin\left(\frac{9}{4}t\right)$$

Step 3: Choose $y(0) = y'(0) = 0$. $c_1 = \frac{16}{17}$ $c_2 = 0$

$$f_\alpha = \frac{\alpha}{2\pi} \quad T_\alpha = \frac{1}{f_\alpha}$$

$$y(t) = \frac{16}{17} \cos(2t) - \frac{16}{17} \cos\left(\frac{9}{4}t\right)$$

$$\alpha = \frac{9}{4} + 2 = \frac{17}{4} \quad T_\alpha = \frac{2\pi}{\alpha} = \frac{16\pi}{17}$$

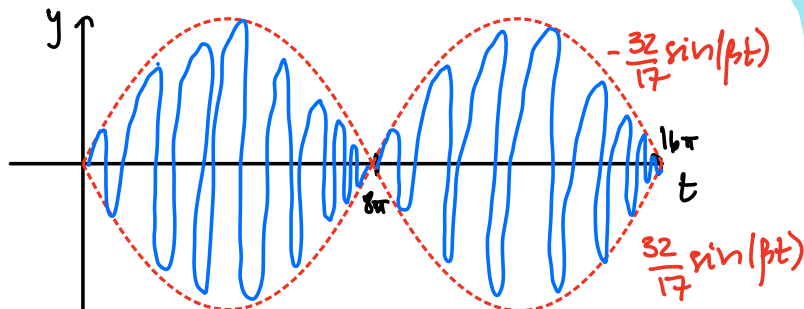
$$y(t) = -2 \cdot \frac{(-16)}{17} \sin(\alpha t) \sin(\beta t)$$

$$\beta = \frac{9}{4} - 2 = \frac{1}{4} \quad T_\beta = \frac{2\pi}{\beta} = 16\pi$$

$$= \frac{32}{17} \sin\left(\frac{17}{8}t\right) \sin\left(\frac{1}{8}t\right)$$

oscillation beating

Sketch:



poll 2.

Wanna check it out on MATLAB?

4.4 Amplitude and Phase of the Steady State.

non-zero damping!

$$y'' + py' + qy = \cos(\omega t) = \operatorname{Re}(e^{i\omega t})$$

Goal: Find $y_p(t) = A \cos(\omega t + \phi)$ steady-state solution.

How? Same way as before $y_c(t) = a e^{i\omega t}$

Subst. into the eq. $y_c' = a(i\omega)e^{i\omega t}$, $y_c'' = a(i\omega)^2 e^{i\omega t} = -a\omega^2 e^{i\omega t}$

$$-a\omega^2 e^{i\omega t} + p \cdot a(i\omega)e^{i\omega t} + q \cdot a e^{i\omega t} = e^{i\omega t}$$

$$-a\omega^2 + p a i \omega + q a = 1$$

$$a(-\omega^2 + p i \omega + q) = 1$$

$$a = \frac{1}{q - \omega^2 + i p \omega} = |a| e^{i\theta}$$

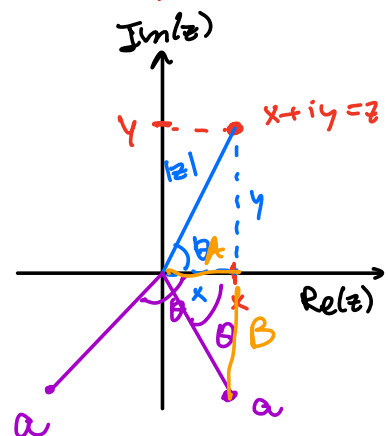
$$|a| = \frac{1}{|q - \omega^2 + i p \omega|} = \frac{1}{\sqrt{(q - \omega^2)^2 + (p\omega)^2}}$$

$$a = \frac{1}{(q - \omega^2) + i p \omega} \cdot \frac{(q - \omega^2) - i p \omega}{(q - \omega^2) - i p \omega} = \frac{q - \omega^2}{\underbrace{(q - \omega^2)^2 + (p\omega)^2}_A} + i \frac{(-p\omega)}{\underbrace{(q - \omega^2)^2 + (p\omega)^2}_B}$$

$$\theta \in [-\pi, 0]$$

$$\arctan\left(\frac{B}{A}\right) = \arctan\left(\frac{-p\omega}{q - \omega^2}\right)$$

$$|x + iy| = \sqrt{x^2 + y^2}$$



p. 428

Amplitude

When is the maximum achieved?

Fix p, q . $A(w) = ((q - w^2)^2 + (pw)^2)^{-1/2}$

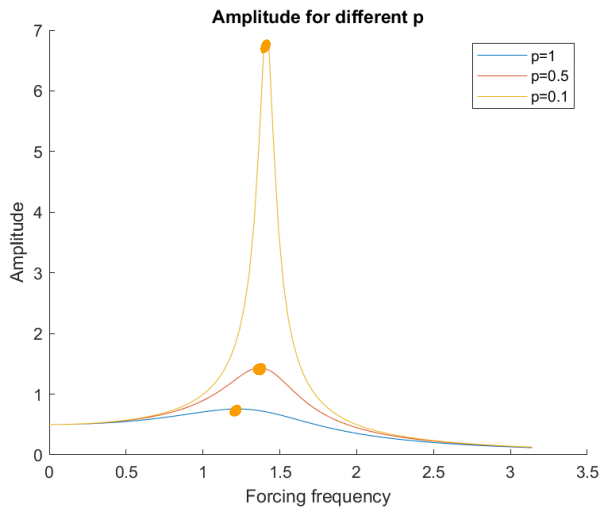
$$A'(w) = -\frac{1}{2} \cdot ((q - w^2)^2 + (pw)^2)^{-3/2} \cdot \left[2(q - w^2) \cdot (-2w) + 2pw \cdot p \right]$$

chain rule

which is zero if and only if is zero.

$$2(q - w^2)(-2w) + 2pw \cdot p = 0 \Rightarrow -2q + 2w^2 + p^2 = 0$$
$$w = \sqrt{q - \frac{p^2}{2}}$$

Amplitude for different dampings.



What can you see here?

$$y'' + py' + qy \approx^{p \approx 0} y'' + qy$$

Section 4.4. for analysis
on Θ .

For a discussion on the phase angle, check p.430.

— End of Chapter 4.

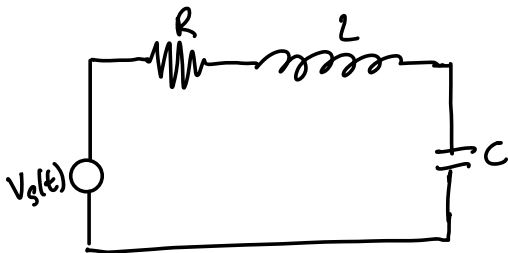
Chapter 5. Nonlinear Systems

The Van der Pol Equation

$$x'' - \mu(1-x^2)x' + x = 0$$

proposed by Van der Pol while studying electric circuits

example of ODEs in circuits: RLC circuit



R: resistor
I: inductor
C: capacitor

$$LCv'' + RCv' + v = V_S(t)$$

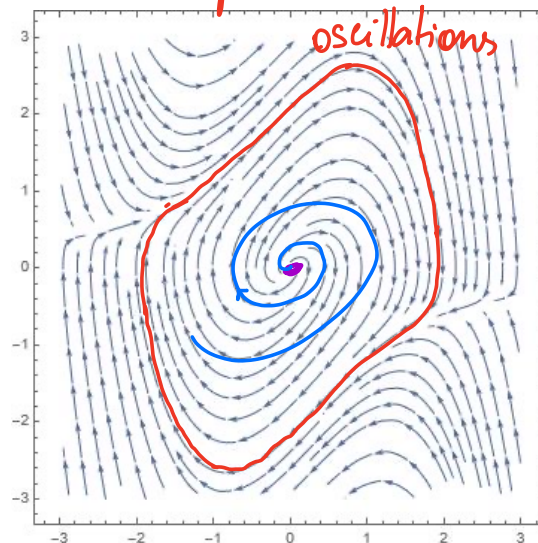
But back to Van der Pol:

$$x' = y$$

$$\mu = 1.$$

$$x' = y$$

$$y' = -x + \mu(1-x^2)y$$



natural questions.

• any equilibrium point?

$$(0,0) \quad \begin{aligned} x' = 0 &\Rightarrow y = 0 \\ y' = 0 &\Rightarrow -x + (1-x^2) \cdot 0 = 0 \\ &\quad x = 0 \end{aligned}$$

• what is the type of the equilibrium point?

$$x' = y$$

$$y' = -x + (1-x^2)y$$

$$x' = y$$

$$y' = -x + y - yx^2 \approx x' = y$$

small x, y $yx^2 \approx 0$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}}_A \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\det(A - \lambda I) = (-\lambda)(1-\lambda) + 1 = \lambda^2 - \lambda + 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{(-1)^2 - 4 \cdot 1}}{2} = \frac{1 \pm i\sqrt{3}}{2}$$

the origin is a spiral source

a) source

b) sink

c) spiral source

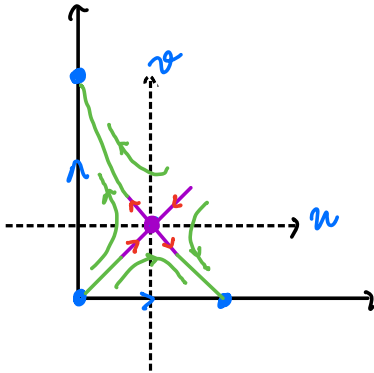
d) spiral sink.

pol 2.

A competing species model

$$x' = 2x(1 - \frac{x}{2}) - xy = x \left[2(1 - \frac{x}{2}) - y \right] = 0$$

$$y' = 3y(1 - \frac{y}{3}) - 2xy = y \left[3(1 - \frac{y}{3}) - 2x \right] = 0$$



(1,1)

$$u = x - 1 \Rightarrow u' = x'$$

$$v = y - 1 \Rightarrow v' = y'$$

$$\begin{aligned} u' &= 2(u+1)(1 - \frac{(u+1)}{2}) - (u+1)(v+1) \\ &= (u+1)(\cancel{1} - (u+1)) - (uv + u + v + 1) \\ &= \cancel{u+1} - u^2 - \cancel{u} - uv - u - v - \cancel{1} \\ &= -u^2 - uv - u - v \end{aligned}$$

$$u' = -u - v - u^2 - uv$$

$$v' = -2u - v - v^2 - 2uv$$

u, v small
 $\approx$

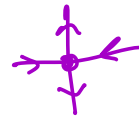
$$\begin{aligned} v' &= (v+1)(\cancel{3} - (v+1)) - 2(u+1)(v+1) \\ &= \cancel{2v+2} - v^2 - v - 2uv - \cancel{2v} - 2u - \cancel{2} \\ &= -v^2 - 2uv - 2u - v. \end{aligned}$$

$$\begin{aligned} u' &= -u - v \\ v' &= -2u - v \end{aligned} = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u' \\ v' \end{bmatrix}$$

eig?

$$\text{tr} A = -2, \det = 1 - 2 = -1.$$

saddle.



A non-polynomial example

$$x' = y$$

$$y' = -y - \sin x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$x' = y$$

$$y' = -y - \sin x = -y - x + \frac{x^3}{3!} - \frac{x^5}{5!} + \dots$$

x, y small

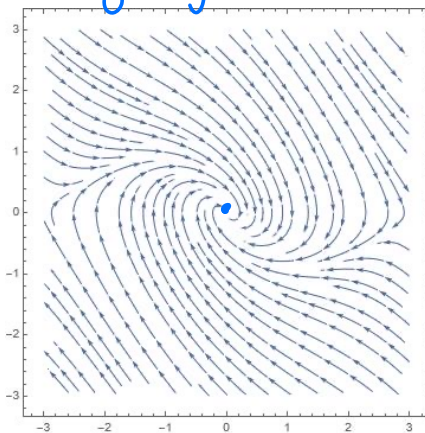
$$\approx x' = y$$

$$y' = -x - y = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

eigenvalues of $\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ are $\lambda^2 + \lambda + 1 = 0$
 $\lambda = \frac{-1 \pm \sqrt{1-4}}{2} \Rightarrow \text{spiral sink.}$

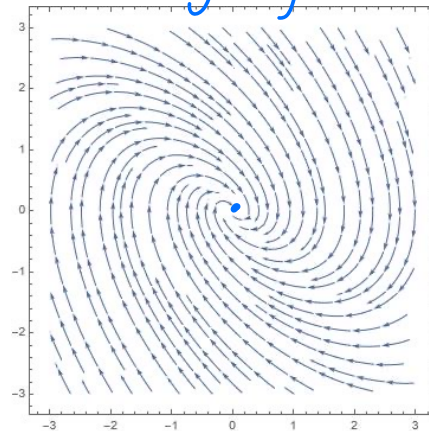
$$x' = y$$

$$y' = -y - \sin(x)$$



$$x' = y$$

$$y' = -y - x$$



the big picture

$$\begin{cases} x' = f(x, y) \\ y' = g(x, y) \end{cases}$$

Suppose that (x_0, y_0) is an equilibrium point, i.e.,

$$\begin{cases} f(x_0, y_0) = 0 \\ g(x_0, y_0) = 0 \end{cases}$$

$$x' = f(x, y) \stackrel{\text{Taylor}}{=} \underbrace{f(x_0, y_0)}_0 + \frac{\partial f}{\partial x}(x_0, y_0) \cdot (x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) + o(2)$$

$$y' = g(x, y) = \underbrace{g(x_0, y_0)}_0 + \frac{\partial g}{\partial x}(x_0, y_0) \cdot (x - x_0) + \frac{\partial g}{\partial y}(x_0, y_0)(y - y_0) + o(2)$$

when x is close x_0 , y is close y_0 , $(x - x_0)^{2,3,\dots} \approx 0$
 $(y - y_0)^2 \approx 0$

$$\begin{aligned} x' = f(x, y) &\approx \begin{bmatrix} \frac{\partial f}{\partial x}(x_0, y_0) & \frac{\partial f}{\partial y}(x_0, y_0) \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} = \underbrace{\begin{bmatrix} \dots \end{bmatrix}}_A \begin{bmatrix} u \\ v \end{bmatrix} \\ y' = g(x, y) &\approx \begin{bmatrix} \frac{\partial g}{\partial x}(x_0, y_0) & \frac{\partial g}{\partial y}(x_0, y_0) \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} \end{aligned}$$

By the homework, it is enough to study

$$A = DF(x_0, y_0)$$

the big picture

"The Jacobian matrix of

$$\text{Let } A = \begin{bmatrix} \frac{\partial f}{\partial x}(x_0, y_0) & \frac{\partial f}{\partial y}(x_0, y_0) \\ \frac{\partial g}{\partial x}(x_0, y_0) & \frac{\partial g}{\partial y}(x_0, y_0) \end{bmatrix}$$

$$F(X) = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix}$$

evaluated at (x_0, y_0) "

Eigenvalues of A :

Real: $\lambda_1, \lambda_2 < 0$
sink

$\lambda_1, \lambda_2 > 0$
source

$\lambda_1 < 0 < \lambda_2$
saddle.

def separatrices.

$$\text{Ex: } \begin{aligned} x' &= -2x + 2x^2 \\ y' &= -3x + y + 3x^2 \end{aligned}$$

equilibria points:

$$x' = 0 \Rightarrow x = 0, 1$$

$$y' = 0 \Rightarrow x = 0 \Rightarrow y = 0$$

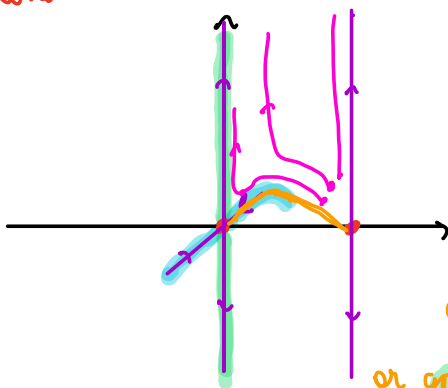
$$x = 1 \Rightarrow y = 0$$

$$A(x, y) = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} = \begin{bmatrix} -2+4x & 0 \\ -3+6x & 1 \end{bmatrix}$$

$$A(0, 0) = \begin{bmatrix} -2 & 0 \\ -3 & 1 \end{bmatrix} \rightarrow \det A = -2 \Rightarrow \text{saddle.}$$

$$A(1, 0) = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$$

when $x=0, x=1 \Rightarrow x'=0$.



Solution that goes to the saddle when $t \rightarrow \infty$
or goes to the saddle when $t \rightarrow -\infty$
are called separatrices.

When the linearization yields complex eigenvalues, you may conclude that:

$\operatorname{Re}(\lambda) < 0$
spiral sink

$\operatorname{Re}(\lambda) > 0$
spiral source

$\operatorname{Re}(\lambda) = 0$
inconclusive.

Ex. $x' = y - (x^2 + y^2)x$ $E_f: (0,0)$

$y' = -x - (x^2 + y^2)y$

$x' = 0 \cdot x + 1 \cdot y - x^3 - xy^2$

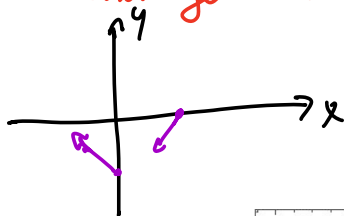
$y' = -1 \cdot x + 0 \cdot y - x^2y - y^3$

$$\begin{aligned} \frac{\partial f}{\partial x} &= -3x^2 - y^2 & \frac{\partial g}{\partial x} &= -1 - 2xy \\ \frac{\partial f}{\partial y} &= 1 - 2xy & \frac{\partial g}{\partial y} &= -x^2 - 3y^2 \end{aligned}$$

Jacobian = $DF(0,0)$

$A(0,0) = \begin{bmatrix} f_x & 0 & f_y & 1 \\ g_x & -1 & g_y & 0 \end{bmatrix} \Rightarrow \lambda^2 + 1 = 0$
 $\lambda = \pm i$

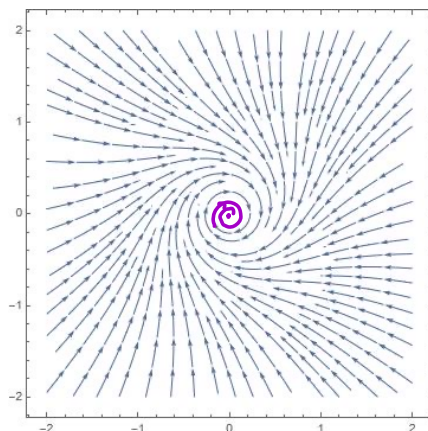
you CAN change variables to POLAR FORM. $(r \cos \theta, r \sin \theta)$



at $(1,0)$, $V(1,0) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

at $(0,-1)$, $V(0,-1) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

center: $V(x,y)$ was



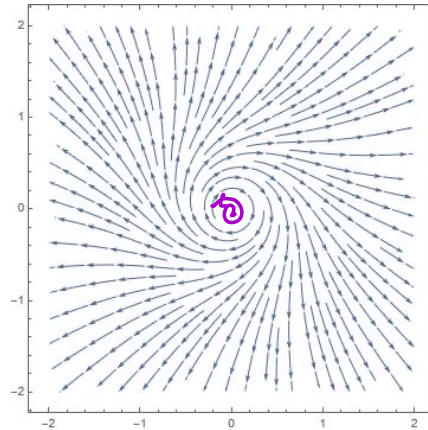
spiral sink.

$\operatorname{eig}(A) = \pm i$

Ex. $x' = y + (x^2 + y^2)x$

$y' = -x + (x^2 + y^2)y$

spiral source.



Conclusion: $A(x_0, y_0)$ & $\operatorname{Re}(\lambda) \geq 0$, you don't have any results.

5.2. Qualitative Analysis.

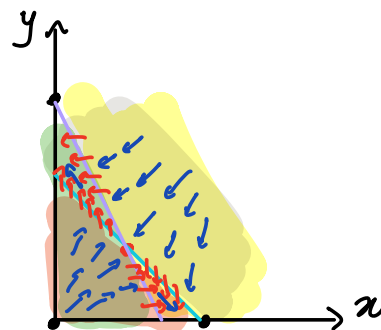
back to competing species.

$x' = 2x(1 - \frac{x}{2}) - xy$ $x=0$
 $y=2-x$

$y' = 3y(1 - \frac{y}{3}) - 2xy$ $y=0$
 $y=3-2x$ $y'=0$

eq. points: $(0,0), (0,3)$
 $(2,0), (1,1)$

↑ saddle.



(we already used this before)

def nullclines.

For a system

$$\begin{cases} x' = f(x, y) \\ y' = g(x, y) \end{cases}$$

the x -nullcline is $\{(x, y) \in \mathbb{R}; f(x, y) = 0\}$

y -nullcline is $\{(x, y) \in \mathbb{R}; g(x, y) = 0\}$

back to competing species.

$$x' = 2x(1 - \frac{x}{2}) - xy$$

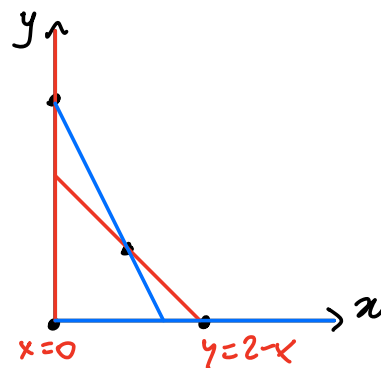
$$y' = 3y(1 - \frac{y}{3}) - 2xy$$

FILL IN

x -nullclines: $x=0$ and $2-x-y=0$

y -nullclines: $y=0$ and $3-2x-y=0$

Claim: you've done this



What is the difference now?

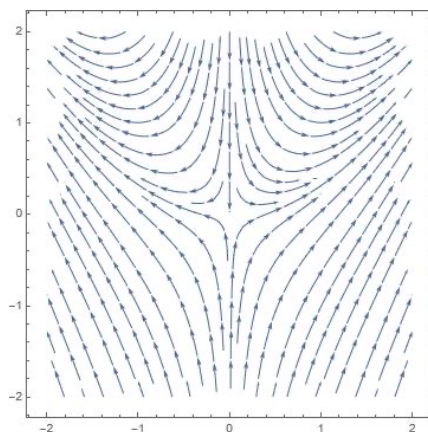
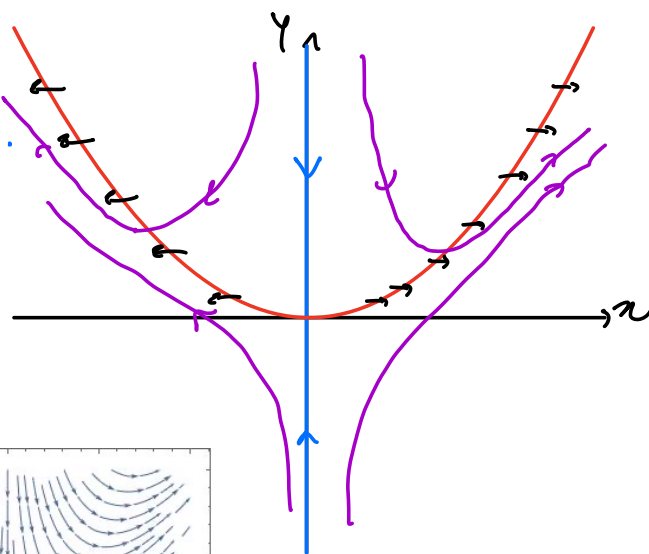
Well, it doesn't have to be straight lines.

Ex. $x' = x$
 $y' = -y + x^2$

x -nullclines: $x' = 0 \Rightarrow x = 0$.

y -nullclines: $y' = 0 \Rightarrow y = x^2$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \text{saddle}$$



Ex

$$x' = x + y - x^3$$

$$y' = -\frac{x}{2}$$

$$x' = 0$$

$$\Rightarrow y = x^3 - x = x(x^2 - 1)$$

$$y' = 0$$

$$\Rightarrow x = 0.$$

